T.B.C. : CDS-MATH-ALL-26-A Serial No. **940280**	**TEST BOOKLET** **[A]** **COMPLETE ARCHIVE**	**Elementary Mathematics** **Marks: +1.000 / -0.333**

UNION PUBLIC SERVICE COMMISSION

COMBINED DEFENCE SERVICES (CDS) EXAMINATION

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EXAMINATION SECTION / CHAPTER	ITEMS	SERIES
Number System & Divisibility	**39 Qs**	Series [A]
HCF, LCM & Logarithms	**9 Qs**	Series [A]
Percentages, Profit, Loss & Discount	**9 Qs**	Series [A]
Simple & Compound Interest, Ratio & Mixtures	**15 Qs**	Series [A]
Time, Speed & Distance, Time & Work	**14 Qs**	Series [A]
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Quadratic Equations & Progressions	**2 Qs**	Series [A]
Trigonometry & Heights and Distances	**2 Qs**	Series [A]
Geometry: Lines, Angles, Triangles & Polygons	**8 Qs**	Series [A]

SECTION : NUMBER SYSTEM & DIVISIBILITY (39 QUESTIONS)

1. A number when divided by 7 leaves a remainder 3 and the resulting quotient when divided by 1 1 leaves a remainder [CDS (I) 2009 Q16]
(a) 35 (b) 40 (c) 45 (d) 50

2. If the same number when divided by 11 leaves a remainder m and the resulting quotient when divided by 7 leaves a remainder n. What are the values of m and n respectively ? [CDS (I) 2009 Q17]
(a) I and (b) 4 and (c) 3 and (d) 6 and 3

3. The number of pairs (x, y) where x, y are integers satisfying the equation 21x+48y=5is [CDS (I) 2009 Q18]
(a) Zero (b) One (c) Two (d) Infinity

4. A is a set of positive integers such that when divided by 2, 3, 4, 5 and the remainder l, 2, and 5 respec- tively. How many integers between 0 and 100 belong to the set A ? [CDS (I) 2009 Q22]
(a) No integer (b) One (c) Two (d) Three

5. What is the least perfect square which is divisible by 3, 4, 5, 6 and 7? [CDS (I) 2009 Q26]
(a) 1764 (b) 17640 (c) 44100 (d) 176400

6. Let N be the greatest number that will divide 600, 631 and 724, leaving the same remainder. What is the value of N? [CDS (I) 2009 Q41]
(a) 41 (b) 37 (c) 33 (d) 31

7. Let N be a 5-digit number. When N is divided by 6, 12, 15, 24 it leaves respectively 2, 8, 11, 20 as remainders. What is the greatest value of N ? [CDS (I) 2009 Q98]
(a) 99960 (b) 99956 (c) 99950 (d) 99946

8. A sum of money compounded annually doubles itself in 5 years. In how many years will it become four times of itself? [CDS (II) 2009 Q12]
(a) 10 years (b) 12 years (c) 15 years (d) 20 years 11

9. Three planets revolve round the sun once in 200, 250, days respectively in their own orbits. When do they all come relatively to the same position as at a certain point of time in their orbits ? [CDS (II) 2009 Q15]
(a) After 3000 days (b) After 2000 days (c) After 1500 days (d) After 1200 days

10. All odd prime numbers upto 110 are multiplied together. What is the unit digit in this product ? [CDS (I) 2010 Q3]
(a) 0 (b) 3 (c) 5 (d) None of the above

11. What are distinct prime mctors of the number 26381 ? [CDS (II) 2010 Q1]
(a) 29, 17, (b) 31, 17, (c) 19, 37, (d) 23, 31,

12. What is the sum of all the divisors of 256 ? [CDS (I) 2011 Q5]
(a) 511 (b) 510 (c) 256 (d) 255 A —

13. If the area of a square is 2401++ 196r2+4, then what is its side length ? [CDS (I) 2011 Q11]
(a) 49x2 -f- 3x + (b) 49x2 _ 3x + (c) 49++2 (d) 59++2

14. What would be the maximum value of Q in the equation 5P9+ 3R7 + 2Q8= 1114? [CDS (II) 2012 Q6]
(a) x = (b) x = (c) 330 (d) 340

15. In an examination, a student was asked to divide a certain number by [CDS (II) 2012 Q7]
(a) x = (b) x = (c) 330 (d) 340

16. By mistake he multiplied it by 8 and got the answer 2016 more than the correct answer. What was, the number ? [CDS (II) 2012 Q8]
(a) 252 (b) 256 (c) 258 (d) 260

17. A man undertakes to do a certain work in 150 days. He employs 200 men. He finds that only a quarter of the work is done in 50 days. How many additional men should he employ so that the whole work is finished in time ? [CDS (II) 2012 Q11]
(a) 75 (b) 85 (c) 100 (d) 120

18. The annual income of a person decreases by T 64 if the annual rate of interest decreases from 4% to 3-75%. What is his original annual income ? [CDS (II) 2012 Q15]
(a) ^ 24,000 (b) r 25,000 (c) ^ 25,600 (d) ^ 24,600

19. The average of the temperatures recorded at noontime from Monday to Sunday is 31 oc. Ifthe lowest tempera- ture recorded is 300C, then what is the maximum of temperature that is possible to record at noontime on any one of the days [CDS (I) 2013 Q4]
(a) 340C (b) ,350C (c) 360C (d) 370C

20. Which one of the following has least number of divisors ? [CDS (II) 2013 Q4]
(a) 88 (b) 91 (c) 96 (d) 99

21. What are the factors of [CDS (I) 2014 Q3]
(a) (x —2y—4) and (x —2Y+2) (b) (x —y and (x —4 y + 4) (c) (x —y and (x —4 y —4) (d) (x (x +2Y+2) 26k /

22. If X takes 45 minutes more than Y to reach the destination, then what is the time taken by Y to reach the destination ? [CDS (I) 2014 Q9]
(a) 135 minutes (b) 150 minutes (c) 180 minutes (d) 225 minutes

23. The price of an article X increases by 20% every year and price of article Y increases by 10% every year. In the year 2010, the price of article X was C 5000 and price of article Y was 0000. In which year the difference in their prices exceeded 0000 for the first time ? [CDS (I) 2015 Q2]
(a) 2012 (b) 2013 (c) 2014 (d) 2015

24. What is the average marks of all the candidates? [CDS (I) 2015 Q6]
(a) 758 (b) 75-8 (c) 760. (d) 768'.

25. What is the sum of their reciprocals ? [CDS (I) 2015 Q9]
(a) 2 (b) 2 (c) 2 (d) 2-562 + b2

— SPACE FOR ROUGH WORK —

26. How many zeros are there in the product 150 x 249 x 348* x 501 ? [CDS (I) 2016 Q7]
(a) 262 (b) 261 (c) 246 (d) 235

27. The number of three digit numbers (all digits are different) which are divisible by 7 and also divisible by 7 on reversing the order of the digits, is [CDS (II) 2016 Q45]
(a) Six (b) Five (c) Four (d) Three

28. What is the sum of the largest and the smallest 4-digit numbers made by using single digit prime numbers (without repetition) ? [CDS (II) 2016 Q58]
(a) 7887 (b) 7997 (c) 8998 (d) 9889

29. If a Star figure is formed by elongating the sides of a regular pentagon, then the measure of each angle at the angular points of the star figure is [CDS (I) 2017 Q12]
(a) 360 (b) 350 (c) 320 (d) 300

30. When the winner wins the race, then the distance between the athlete in the second position to the athlete in the third mysition is [CDS (I) 2018 Q4]
(a) 1000m (b) 800m (c) 750m (d) 600 m

31. If the average of 9 consecutive positive integers is 55, then what is the largest integer ? [CDS (I) 2019 Q100]
(a) 57 (b) 58 (c) 59 (d) 60

32. If the average marks of the complete class is 72, then what is the average marks of the girls ? [CDS (I) 2020 Q11]
(a) 64 (b) 65 (c) 68 (d) 74

33. A cricketer has a certain average of 10 innings. In the eleventh inning he scored 108 runs, thereby increasing his average by 6 runs. What is his new average ? [CDS (II) 2021 Q4]
(a) 42 (b) 47 (c) 48 (d) 60

34. A wheel makes 360 revolutions in one minute. What is the number of radians it tums in one second ? [CDS (II) 2021 Q9]
(a) 10 (b) 16i (c) 36 (d) 27

35. Two circles touch internally. The sum of their areas is 1367t cm2 and distance between their centres is 4 What are the [CDS (I) 2022 Q4]
(a) +1 (b) + (c) l (d) 266+1

36. The area of a rectangle whose length is 5 units more than twice its width is 75 square units. What is its width ? [CDS (I) 2022 Q9]
(a) 3 units (b) 5 units (c) 7 units (d) 10 units

37. Which product is produced least over the years ? [CDS (II) 2023 Q10]
(a) Type (b) Type (c) Type (d) Type V ,'SOA

38. In Which one of the following pairs of years, the difference in total number of tablets produced between them is m inimum? [CDS (II) 2023 Q11]
(a) (2003, 2005) (b) (2001, 2005) (c) g003, 2004) (d) (2000, 2002)

39. Ifthe length of the shadow of a tower is equal to its height, then what is the Sun's altitude at that time ? [CDS (II) 2025 Q10]
(a) 150 (b) 300 (c) 450 (d) 600

SECTION : HCF, LCM & LOGARITHMS (9 QUESTIONS)

40. If the highest common factor of two positive integers is 24 then their least Common multiple cannot [CDS (II) 2010 Q65]
(a) 72 (b) 216 (c) 372 (d) 600

41. The number of digits in 330 is n and it is given that log103 -04771. What is the value of n ? [CDS (II) 2010 Q96]
(a) 13 (b) 14 (c) 15 (d) 16 - A

42. There are two numbers which are greater than 21 and their LCM and HCF are 3003 and 21 respectively. What is the sum of these numbers ? [CDS (I) 2011 Q96]
(a) 504 (b) 508 (c) 514 (d) 528

43. A can do 50% more work than B in the same time. B alone can do a piece of work in 30 hours. B starts working and had already worked for 12 hours when A joins him. How many hours should B and A work together to complete the remaining work ? [CDS (II) 2013 Q19]
(a) 6 hours (b) 12 hours (c) hours (d) 7-2 hours

44. men are supposed to complete a work in 10 days. Aner working for 5 days, they realise that only one-fourth of the work is done. How many more men they need to employ to finish the work on time ? [CDS (II) 2014 Q17]
(a) 40 (b) 30 (c) 20 (d) 15

45. What is the greatest possible speed at which a person can walk 13-3 km and 209 km so that the time (in hours) required in each case is a whole number ? [CDS (II) 2014 Q22]
(a) 1-7 km/h (b) 1-9 km/h (c) km/h (d) 2•3 km/h

46. From a rectangular metal sheet of sides 25 cm and 20 em, a circular sheet as large as possible is cut-off. What is the area of the remaining sheet ? [CDS (I) 2018 Q17]
(a) 186 (b) 144 (c) 93 cm2 (d) 72 cm2

47. ABCD is a square. If the sides AB and CD are increased by 30%, sides BC and AD are increased by 20%, then the area of the resulting rectangle exceeds the area of the square by (b) 52% (c) 540/0 (d) 56% [CDS (I) 2019 Q18]
(a) will be 75. Let (b) 52% (c) 540/0 (d) 98 cm 108 cm2

48. Ihe present age of a father is equal to sum of the ages of his 4 children. Aner ten years the sum of the ages of the children will be 1-6 times the age of their father. What is the present age of father ? [CDS (II) 2025 Q19]
(a) 36 years (b) 40 years (c) 42 years (d) 45 years

— SPACE FOR ROUGH WORK —

SECTION : PERCENTAGES, PROFIT, LOSS & DISCOUNT (9 QUESTIONS)

49. There are three brothers. The sums of ages of two of them at a time are 4 years, 6 years and 8 years. The age difference between the eldest and the youngest is [CDS (II) 2009 Q25]
(a) 3 years (b) 4 years (c) 5 years (d) 6 years

50. A 4-digit number N has exactly 15 distinct divisors. What is the total number of distinct divisors of ? [CDS (II) 2009 Q26]
(a) 16 (b) 30 (c) 45 (d) 225

51. The minute hand of a clock is 15 cm long and sweeps an area of 15" cm on the dial of the clock. How much angle does it describe during this period ? [CDS (I) 2011 Q23]
(a) 120 (b) 180 (c) 240 (d) 300

52. There are two categories of employees whose strength increased by same percentage in 2020. What are the two categories ? [CDS (I) 2012 Q18]
(a) E and F (b) C and E (c) A and E (d) and C A -

53. In which year was the percentage in- crease in production of cars minimum as compared to its previous year ? [CDS (I) 2012 Q19]
(a) 2015 (b) 2016 (c) 2017 (d) 2018

54. What is the percentage of zinc in X ? [CDS (I) 2012 Q21]
(a) 400/0 (b) 36% (c) 34% (d) 30% A -

55. Number of credit cards held by an Indlvldual can be treated as [CDS (I) 2012 Q27]
(a) Qualitative data (b) Discrete data (c) Categorical data (d) None of the above

56. men working 8 hours a day can completely build a wall of length 100 m, breadth 20 cm and height 5 m in 10 days. How many days will 16 men working 10 hours a day require to build a wall of length 200 m, breadth 60 cm and height 5 m ? [CDS (II) 2015 Q24]
(a) 18 (b) 24 (c) 27 (d) 36

57. What is the sum of the radii of the circles ? [CDS (II) 2024 Q29]
(a) 104 cm (b) 11-2 cm (c) 12 cm (d) 13 cm

SECTION : SIMPLE & COMPOUND INTEREST, RATIO & MIXTURES (15 QUESTIONS)

58. Which one of the following numbers is not a square of any natural ? [CDS (II) 2009 Q38]
(a) 5041 (b) 9852 (c) 1936 (d) 6241 T-DETA.K.FN - A

59. If 6 men and 8 boys can do a piece of work in 10 days while 26 men and 48 boys can do the sarru in 2 days, what is the time taken by 15 men and 20 boys in doing the same type of work ? [CDS (II) 2011 Q36]
(a) 4 days (b) 5 days (c) 6 days (d) 7 days

60. There are two employees X and Y. X's salary is first increased by 12% and then decreased by 10%, and Y's salary is first increased by 10% and then decreased by 12%. If their salaries at present are equal, then what was the ratio of initial salary of X to initial salary of Y? [CDS (II) 2012 Q96]
(a) 50: (b) 51: (c) 121: (d) 121: 125 I P.T.O.

61. The wages, of labourers in a factory has increased in the ratio 22 : 25 and their number decreased in the ratio 3 : [CDS (II) 2012 Q97]
(a) 4 days (b) 5 days (c) 6 days (d) days 33. If of

62. 10% annum on compound interest for 3 years, compounded annually, and paid back in 3 equal annual installments during these years. What is the amount of each installment (approximately) ? [CDS (I) 2013 Q1]
(a) e 4,437 (b) e 4,237 (c) r 4,021 (d) r 3,811

63. In what duration of time will *3300 become *3399 at 12% per annum interest compounded quarterly ? [CDS (I) 2013 Q2]
(a) 3 months (b) 6 months (c) 9 months (d) 1 year

64. TWO men, and B run a 4 km racc on a course km round. If their speeds are in the ratio 5 : 4, how often does the winner pass the other ? [CDS (II) 2013 Q89]
(a) Once (b) Twice (c) Thrice (d) Four times - A

65. Area of APDA is equal to that of APCQ [CDS (II) 2014 Q37]
(a) — AB (b) a AB2 (c) 1, 2,3 82. 83. (d) 4

66. cm What is the approximate area of the shaded region in the figure given ? [CDS (II) 2014 Q38]
(a) 15•3 cm2 (b) 25•5 cm2 (c) 284 cm2 (d) cm2

67. What is the measure of an interior angle of polygon ? [CDS (I) 2015 Q33]
(a) 1100 (b) 1200 (c) 1300 (d) 1400

68. Sex ratio is defined as the number of females per 1000 males. In a place the total inhabitants are out of which 935,000 are females. What is the sex ratio for the place ? [CDS (II) 2016 Q73]
(a) 935 (b) 1000 (c) 1935 (d) 9350 6 2 23

69. The length and breadth of a rectangle are in the ratio 4 : [CDS (II) 2016 Q76]
(a) — •E (b) litre (c) litre , (d) — litre

70. What is the maximum area of a rectangle, the IVimeter of which is 18cm? [CDS (II) 2017 Q34]
(a) 2025 cm2 (b) 20-00 cm2 (c) 19-75 cm2 (d) 19-60 cm2

71. In a pi-diagram there are three sectors. If the ratio of the angles of the sectors is 1 : 2 : 3, then what is the angle of the largest sector ? [CDS (II) 2017 Q45]
(a) 2000 (b) 1800 (c) 1500 (d) 1200

— SPACE FOR ROUGH WORK —

72. A tank can be filled by pipe X in 2 hours and pipe Y in 6 hours. At 10 a.m. pipe X was opened. At what time will the tank be filled if pipe Y is opened at 11 a.m. ? [CDS (I) 2019 Q37]

(a) 12:45 hours (b) 5:00 p.m. (c) 11:45 a.m. (d) 11:50 a.m.

SECTION : TIME, SPEED & DISTANCE, TIME & WORK (14 QUESTIONS)

73. A river 2-5m deep and 45m wide is flowing at the speed of km/hour. The amount of water that runs into the sea per minute is [CDS (II) 2013 Q92]

(a) 6650 (b) 6750 (c) 6850 (d) 6950 - A

74. A person can row downstream 20 km in 2 hours and upstream 4 km in 2 hours. What is the speed of the current ? [CDS (II) 2013 Q93]

(a) 2 km/hour (b) 2-5 km,'hour (c) 3 km/hOur (d) 4 knvhour

75. A motorist travels to a place 150 km away at an average speed of 50 km/hour and returns at 30 km/hour. What is the average speed for the whole journey ? [CDS (II) 2013 Q94]

(a) 35 km,fhour (b) 37 km,fhour (c) 37-5 (d) 40 km/hour

76. A car travels from A to B at a speed of 40 km/hr, travels back from B to A at a speed of 30 km/hr and again goes from A to B at a speed of 60 km/hr. What is the average speed of the car? [CDS (II) 2013 Q97]

(a) km/hr (b) 42 km/hr (c) 40 km/hr (d) km/hr

77. A train takes two hours less for a journey of 300 km if its speed is increased by 5 knvhr from its usual speed. What is its usual speed ? [CDS (I) 2014 Q31]

(a) 50 knvhr (b) 40 km/hr (c) 35 knvhr (d) 25 km/hr

78. There are some benches in a class room having the number of rows 4 more than the number of columns. If each bench is seated with 5 students, there are two seats vacant in a class of 158 students. The number of rows is (d) [CDS (II) 2014 Q45]

(a) ac I If f (b) -20 (c) 20 (d) 12

79. Two students A and B solve an equation of the form ± px + q = O. A starts with a %T0ng value of p and obtains the roots as 2 and [CDS (I) 2015 Q44]

(a) p (b) 3 and —4 (c) —3 and—4 (d) —3 and

80. By increasing the speed of his car by 15 km/hour, a person covers 300 km distance by taking an hour less than before. The original speed of the car [CDS (II) 2016 Q80]

(a) 45 km/hour (b) 50 km/hour (c) 60 kni/hour (d) 75 km/hour

81. A man walking at 5 km/hour noticed that a 225 m long train coming in the opposite direction crossed him in 9 seconds. The speed of the train is [CDS (II) 2016 Q83]

(a) 75 km/hour (b) 80 km / hour (c) 85 km / hour (d) 90 kin/ hour w/ 54A

82. When the speed 0fa train is increased by 20%, it takes 20 minutes less to cover the same distance. What is the time taken to cover the same distance with the original speed ? [CDS (II) 2016 Q86]

(a) 140 minutes (b) 120 minutes (c) 100 minutes (d) 80 minutes

83. A person rode one-third of a journey at 60 km/hr, one-third at 50 km/hr and the rest at 40 km/hr. Had the person ridden half of the journey at krnfhr and the rest at 40 km/hr, he would have taken 4 minutes longer to complete the journey. What distance did the person ride ? [CDS (II) 2016 Q90]

(a) 180 km (b) 210 km (c) 240 km (d) 300 km

84. Without stoppages, a train travels at an average speed of 60 km/h and with stoppages it covers the same distance at an average speed of 48 km/h. HOW many minutes per hour does the train stop ? [CDS (II) 2016 Q95]

(a) 10 minutes (b) 12 minutes (c) 1 S minutes (d) 20 minutes

85. A train moving with a speed of 60 km per hour crosses an electric pole in 30 seconds. What is the length of the train in metres ? [CDS (I) 2018 Q27]

(a) 300 (b) 400 (c) 500 (d) 600

86. 31-5 km/h 365 km/h 38-5 krn/h 40-5 km/h c In the figure given above, the side of square ABCD is 7 cm. What is the area of the shaded portion, formed by the arcs of the circles with centre at C and A ? [CDS (II) 2023 Q82]

(a) 7 crn2 (b) 28 cm2 (c) 14 cm2 (d) 21 Crn2 - A

SECTION : ALGEBRA, POLYNOMIALS & LINEAR EQUATIONS (2 QUESTIONS)

87. Which one of the following is a zero of the polynomial 3x^ + 4x^ - 7 ? [CDS (II) 2015 Q10]

(a) 0 (b) 1 (c) 2 (d) -

88. What is the number of students in the class ? [CDS (II) 2024 Q53]

(a) 100 (b) 105 (c) 110 (d) 120

SECTION : QUADRATIC EQUATIONS & PROGRESSIONS (2 QUESTIONS)

89. If a and are the roots of the equation then what is equal to? [CDS (I) 2016 Q18]

(a) 16 (b) 25 (c) 36 (d) 49 1 P.T.O.

90. A quadratic polynomial ax^ + bx + c is such that when it is divided by x, (x - 1) and (x + 1), the remainders are 3, 6 and 4 respectively. What is the value of (a + b) ? [CDS (I) 2016 Q70]

(a) 3 (b) 2 (c) 1 (d) -

SECTION : TRIGONOMETRY & HEIGHTS AND DISTANCES (2 QUESTIONS)

— SPACE FOR ROUGH WORK —

91. ABC is a right triangle with right angle at A. If the value of tan B = any.real k the length of the hypotenuse is of the form (b) 2k (d) 9k (c) 5k [CDS (II) 2017 Q12]

(a) 4 days (b) 5 days (c) 6 days (d) days 33. If of

92. A pole stands vertically inside a triangular park ABC. If the angle of elevation of the top of the pole from each corner of the park is same, then in the triangle ABC, the foot of the pole is at the [CDS (II) 2017 Q15]

(a) Centroid (b) Circumference (c) Incentre (d) Orthocentre

SECTION : GEOMETRY: LINES, ANGLES, TRIANGLES & POLYGONS (8 QUESTIONS)

93. ABC is a triangle right-angled at B. Points M and N trisect BC. If AM = 6 cm and AN = 9 cm, then what is AC equal to ? [CDS (II) 2017 Q21]

(a) 4xfiö cm (b) 265 cm (c) 24 cm (d) 20 cm

94. E is the midpoint of the median AD of a triangle ABC If BE produced meets the side AC at F, then CF is equal to [CDS (I) 2022 Q73]

(a) AC/3 (b) 2AC/3 (c) AC/2 (d) None of the above

95. In an equilateral triangle another equilateral triangle is drawn inside joining the mid-points of the sides of given equilateral triangle and the process is continued up to 7 times. What is the ratio of area of fourth triangle to that of seventh triangle ? [CDS (I) 2022 Q77]

(a) 256 : I (b) 128 : (c) 64: (d) 16:1

96. What is the area of a triangle with sides of length 12 cm, 13 cm and 5 cm ? [CDS (I) 2022 Q79]

(a) 30 crn2 (b) 35 cm2 (c) 40 cm2 (d) 42

97. If the angles of a triangle are 300 and 450 and the included side is + 1) cm, then what is the area of the triangle ? [CDS (I) 2022 Q81]

(a) +1) cm2 (b) + 3) cm2 (c) l) cm2 (d) 266+1) cm2

98. The lengths of sides of a triangle are where If one of the angles is 900, then what are the minimum integral values of x, y, z respectively ? [CDS (I) 2022 Q83]

(a) — AB (b) a AB2 (c) 1, 2,3 82. 83. (d) 4

99. In , a right triangle ABC, BD is perpendicular on hypotenuse AC If AC 9 cm and AD 4 cm, then what is AB+ BC approximately equal to? [CDS (I) 2022 Q87]

(a) 12 cm (b) 12•2 cm (c) 12•4 cm (d) 126 cm

100. The base of an isosceles triangle is 300 units and each of its equal sides is 170 units. Then the area of the triangle is [CDS (I) 2022 Q100]

(a) 9600 square units (b) 10000 square units (c) 12000 square units (d) None of the above

— SPACE FOR ROUGH WORK —